

Table 2: Component scores of the 70 candidate UAV-ISAC research topics.

Research Topic	Signif. (20%)	Novelty (15%)	PHY (20%)	Closure (15%)	Practical (10%)	Literature (10%)	Scope (5%)	Execution (5%)	Total
Near-Field ISAC for Six-Dimensional UAV Pose Estimation During Landing	9.90	9.89	9.90	9.89	9.87	9.89	9.89	9.25	9.86
High-Speed UAV Sensing With 5G NR Signals: Ambiguity Analysis and Pilot Design	9.82	9.80	9.86	9.80	9.86	9.78	9.85	9.75	9.82
Quickest UAV Detection With Adaptive ISAC Beam Sweeping	9.81	9.76	9.75	9.80	9.77	9.76	9.79	10.00	9.79
Recovering UAV Micro-Doppler From Punctured OFDM Signals	9.77	9.74	9.74	9.75	9.76	9.80	9.76	9.81	9.76
Noncoherent Cell-Free ISAC for UAV Detection	9.77	9.77	9.77	9.73	9.74	9.70	9.71	9.35	9.73
UAV Uplink Signals for Passive Bistatic Airspace Sensing	9.75	9.75	9.72	9.74	9.70	9.73	9.70	9.29	9.71
Soft-Information-Aided UAV Detection With OFDM Signals	9.73	9.66	9.68	9.68	9.73	9.70	9.68	9.60	9.69
Numerology Selection for High-Mobility UAV ISAC	9.67	9.68	9.65	9.65	9.67	9.70	9.62	9.57	9.66
Coherent UAV Detection Under Range-Doppler Migration	9.62	9.67	9.63	9.64	9.67	9.68	9.60	9.57	9.64
Sensing-Assisted Beam Alignment for UAV Relaying With Information Causality	9.62	9.62	9.64	9.64	9.58	9.61	9.60	9.40	9.61
Ghost-Target Suppression in Multi-TRP ISAC for Multi-UAV Surveillance	9.54	9.60	9.55	9.59	9.62	9.60	9.59	9.64	9.58
Doppler-Resilient PRS Design for UAV Localization	9.58	9.58	9.58	9.58	9.52	9.51	9.53	9.29	9.55
Phase-Noise-Limited OFDM Sensing of High-Speed UAVs	9.49	9.49	9.56	9.54	9.52	9.56	9.51	9.64	9.53
UAV Data Collection With Joint Device Localization and Clock Calibration	9.50	9.51	9.50	9.51	9.46	9.52	9.52	9.46	9.50
Coherent Processing Interval Design for UAV-Borne ISAC Under Attitude Jitter	9.45	9.50	9.49	9.45	9.47	9.49	9.51	9.36	9.47
Three-Dimensional UAV Localization With Asynchronous Multi-TRP ISAC	9.41	9.46	9.43	9.48	9.44	9.47	9.46	9.54	9.45
Separating Doppler and Carrier Offset in Reciprocal UAV ISAC	9.38	9.38	9.42	9.38	9.38	9.40	9.43	9.93	9.42
Sensing-Aided Beam Tracking Under Fast UAV Attitude Variations	9.44	9.43	9.36	9.41	9.43	9.41	9.40	9.20	9.40
Measurement Scheduling for Cell-Free UAV Tracking	9.34	9.40	9.40	9.41	9.38	9.41	9.42	9.21	9.38
UAV-Enabled Backscatter ISAC for Batteryless Tag Localization	9.33	9.32	9.33	9.38	9.37	9.34	9.39	9.65	9.36
Multi-Band ISAC for Unambiguous UAV Tracking	9.37	9.30	9.31	9.38	9.34	9.37	9.31	9.31	9.34
Discriminative ISAC Waveform Design for UAV-Bird Recognition	9.28	9.36	9.36	9.29	9.34	9.28	9.36	9.29	9.32
Rate-Constrained Sensing-Statistic Feedback for UAV Target Tracking	9.34	9.33	9.26	9.29	9.26	9.32	9.33	9.25	9.30
Multi-TRP UAV Detection With Unknown Bistatic Reflectivities	9.25	9.25	9.24	9.24	9.26	9.31	9.24	9.79	9.28
Distributed Micro-Doppler Fusion for UAV Classification	9.28	9.29	9.27	9.25	9.26	9.25	9.28	9.08	9.26
Over-the-Air Phase Calibration for Cooperative UAV Sensing	9.20	9.26	9.23	9.21	9.26	9.28	9.21	9.38	9.24
Near-Field Beam Tracking for Low-Altitude UAV Communications	9.18	9.20	9.21	9.20	9.19	9.21	9.22	9.62	9.22
OTFS Sensing of Accelerating UAVs	9.24	9.21	9.16	9.18	9.16	9.18	9.24	9.31	9.20
Echo-Assisted Physical-Layer Authentication for Cellular UAVs	9.17	9.22	9.19	9.21	9.22	9.18	9.15	8.92	9.18
Sensing-Aided Grant-Free Access for Cellular-Connected UAVs	9.20	9.20	9.20	9.15	9.14	9.17	9.16	8.77	9.16
Finite-Blocklength ISAC for Timely UAV Detection	9.11	9.13	9.17	9.12	9.14	9.14	9.11	9.26	9.14
Sensing-Assisted Beam Switching Under Air-to-Ground Blockage	9.10	9.13	9.08	9.12	9.15	9.09	9.09	9.36	9.12
Sensing-Aided Robust Secrecy for UAV Communications With a Mobile Eavesdropper	9.11	9.07	9.14	9.08	9.14	9.13	9.06	8.95	9.10
Joint Trajectory and Waveform Design for UAV-Borne Target Tracking	9.12	9.09	9.11	9.11	9.10	9.04	9.12	8.68	9.08
Rotor-Speed Estimation From Cellular ISAC Echoes	9.09	9.04	9.10	9.10	9.04	9.04	9.05	8.81	9.06
Two-Way ISAC for Clock-Offset-Invariant UAV Swarm Localization	9.03	9.00	9.06	9.06	9.07	9.08	9.00	8.96	9.04
UAV-Borne Passive ISAC With Cellular Downlink Signals	9.00	8.99	9.02	8.99	9.01	9.05	9.05	9.21	9.02
UAV Illumination for Bistatic ISAC: Geometry and Power Allocation	8.97	9.04	8.99	8.98	8.97	9.03	9.03	9.07	9.00
Wideband Beam Tracking for UAV ISAC Under Beam Squint	9.00	8.97	8.97	8.97	8.99	8.96	8.97	9.03	8.98
Superimposed Pilots for UAV Sensing and Communications	8.99	8.95	8.99	8.96	9.00	8.97	8.98	8.63	8.96
Sensing With HARQ Retransmissions for Cellular-Connected UAVs	8.98	8.98	8.92	8.94	8.94	8.91	8.95	8.79	8.94
Active Sensing for UAV Jitter Tracking	8.92	8.90	8.89	8.90	8.93	8.90	8.96	9.14	8.92
UAV Detection With Unknown Noise Power in OFDM ISAC	8.87	8.91	8.87	8.92	8.92	8.94	8.88	8.95	8.90
Bistatic Diversity for UAV Detection Under Swerling Fading	8.90	8.88	8.91	8.84	8.84	8.84	8.86	8.98	8.88
Inter-Cell Coordination for UAV Detection in ISAC Networks	8.89	8.88	8.85	8.85	8.88	8.82	8.89	8.76	8.86
Task-Oriented Fronthaul Quantization for Cell-Free UAV Localization	8.84	8.82	8.85	8.82	8.88	8.81	8.87	8.87	8.84
One-Bit Fronthaul for Cell-Free UAV Detection	8.84	8.82	8.86	8.79	8.85	8.85	8.85	8.52	8.82
AFDM Sensing of UAVs With Doppler Spread	8.84	8.78	8.80	8.84	8.76	8.83	8.83	8.57	8.80
Low-Resolution ADCs for Bistatic UAV Sensing	8.78	8.79	8.81	8.80	8.79	8.82	8.76	8.49	8.78
Full-Duplex UAV ISAC With Residual Self-Interference	8.75	8.73	8.77	8.75	8.79	8.74	8.76	8.86	8.76
Constant-Envelope Waveform Design for UAV ISAC	8.73	8.77	8.71	8.73	8.73	8.72	8.73	8.91	8.74
Bistatic ISAC With a UAV Relay Under Direct-Link Interference	8.68	8.72	8.69	8.70	8.71	8.71	8.69	9.13	8.72
Mobility-Assisted Airspace Sensing in Cellular UAV Networks	8.71	8.73	8.67	8.66	8.68	8.69	8.69	8.88	8.70
Sensing-Dwell Placement for UAV-Borne ISAC With Moving Targets	8.65	8.70	8.71	8.72	8.66	8.70	8.64	8.54	8.68

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Research Topic	Signif. (20%)	Novelty (15%)	PHY (20%)	Closure (15%)	Practical (10%)	Literature (10%)	Scope (5%)	Execution (5%)	Total
Synchronization-Aware Role Assignment for Multistatic UAV Sensing	8.64	8.70	8.68	8.62	8.64	8.66	8.67	8.69	8.66
Transmit-Receive Mode Selection for Multi-UAV Bistatic ISAC	8.62	8.63	8.62	8.61	8.64	8.60	8.64	9.00	8.64
Coherent UAV-Swarm ISAC With Imperfect Relative Positioning	8.59	8.66	8.63	8.63	8.66	8.63	8.60	8.47	8.62
Event-Triggered Sensing for UAV Beam Tracking	8.59	8.63	8.62	8.62	8.59	8.59	8.58	8.47	8.60
Robust UAV Communications With Sensing-Dependent Position Errors	8.62	8.62	8.62	8.56	8.58	8.55	8.60	8.24	8.58
Sparse UAV Formation Design for Distributed ISAC	8.55	8.59	8.53	8.55	8.53	8.60	8.56	8.64	8.56
Orientation-Robust Beamforming With Conformal UAV Arrays	8.54	8.55	8.57	8.52	8.54	8.51	8.53	8.52	8.54
Covert UAV Communications Under Public ISAC Signaling	8.49	8.51	8.56	8.51	8.49	8.52	8.53	8.59	8.52
Double-Pass RIS Beamforming for UAV-Borne ISAC	8.51	8.48	8.51	8.53	8.50	8.48	8.47	8.46	8.50
Doppler-Aware Aerial RIS for Terrestrial ISAC	8.44	8.44	8.49	8.49	8.50	8.45	8.44	8.75	8.48
User-Centric Clustering for Cellular UAV Communication and Localization	8.50	8.50	8.47	8.47	8.50	8.43	8.42	8.13	8.46
Information-Causal ISAC for UAV Base Stations With Wireless Backhaul	8.44	8.40	8.46	8.40	8.45	8.48	8.44	8.50	8.44
True-Time-Delay Beamforming for Wideband UAV ISAC	8.41	8.42	8.40	8.41	8.45	8.45	8.45	8.42	8.42
Pointing-Robust FSO Backhaul for UAV-Borne ISAC	8.40	8.41	8.42	8.36	8.44	8.44	8.42	8.23	8.40
Adaptive Beam Probing for UAV ISAC Under Markov Blockage	8.42	8.35	8.34	8.40	8.36	8.40	8.36	8.43	8.38
Cache-Aware UAV Multicast ISAC With Sensing-Freshness Constraints	8.21	8.19	8.18	8.23	8.19	8.22	8.18	8.18	8.20